

Tanush Yarram

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EXPERIENCE

Software Engineer | Visa Inc

March 2026 - Present

- Assumed end-to-end ownership of a production NGINX API gateway fronting internal services, hardening the request path with Message-Level Encryption (MLE) and mutual TLS to encrypt payloads in transit and enforce certificate-based service identity for every consumer
- Led the gateway's security posture, triaging vulnerability scan findings and driving NGINX and dependency patching to remediation via zero-downtime rolling upgrades with no disruption to downstream consumers
- Served as gateway subject-matter expert for partner teams across the org, designing their routing, TLS, and certificate configurations and authoring onboarding runbooks that turned repeat integration requests into a self-service path
- Reduced chatbot platform response times to sub-200ms by building Spring Boot microservices with circuit breakers, Redis caching, and async Kafka event processing, and tuning HikariCP connection pooling across 3 services to cut query latency 15% at 99.9% SLA
- Engineered Grafana observability dashboards with automated threshold alerting across 5 production services, cutting mean-time-to-detect 20%, and authored automated integration test suites that reduced release-blocking failures 40% and enabled weekly independent deploys

Software Engineer Intern | Visa Inc

May - August 2025

- Migrated 8 legacy monolithic applications to Docker/Kubernetes, reducing cold-start time from 4+ minutes to under 30 seconds and eliminating ~\$200K/year in VM infrastructure costs, later adopted as the team's standard deployment pattern
- Identified and resolved CPU and memory bottlenecks across 6 services during cross-departmental code reviews, improving application responsiveness and reducing p99 execution latency by ~55% in the highest-traffic service paths

Software Engineer Intern | Diasorin

April 2024 - August 2024

- Engineered automated data validation pipelines using Python Pandas and SQL to clean and process 20k+ row enterprise datasets, eliminating critical reporting errors and ensuring data integrity
- Developed Java/Spring Boot backend services processing daily ETL workflows for 3 downstream BI teams, replacing manual handoffs and reducing reporting pipeline errors by ~30% with unit and integration test coverage deployed via Azure DevOps CI/CD pipelines

PROJECTS

Momentum Autonomous Trading Agent + Pre-Trade Risk Analysis Dashboard | Next.js, Python, SQLite, Docker

2025 - Present

- Built an end-to-end agentic trading system: an autonomous LLM agent analyzing live market data and executing long-only equity trades via Model Context Protocol (MCP) brokerage integrations on a zero-intervention loop, plus a full-stack pre-trade portfolio impact app
- Constrained agent autonomy within a 15% maximum-drawdown limit via layered risk controls: ATR-based position sizing, correlation filtering, and a tiered circuit breaker with git-backed state persistence for auditable decisions
- Implemented stateful financial algorithms in the analysis layer, cross-account wash-sale detection over rolling 30-day windows, FIFO lot selection, and configurable sector-concentration scoring, validated by a Vitest suite covering tax and risk logic

Auto-Aim Optimization for Embedded Systems | Python, C++, OpenCV, Kalman/Particle Filters

August 2025

- Designed and open-sourced a hybrid Particle-Kalman algorithm fusing vision and IMU sensor data to achieve real-time localization on a single NVIDIA Jetson — replacing UColorado's reference algorithm that exceeded the compute budget by 2×
- Reduced average compute load by 40% versus existing implementations while maintaining targeting accuracy, enabling deployment on embedded hardware previously considered underpowered for the task

AI-Powered Career Guidance Platform | ReactJS, NodeJS, PostgreSQL, OpenAI API, Docker

January 2025

- Cut LLM API latency ~40% and reduced token costs by architecting a RAG pipeline (LangChain + GPT-4) with Redis response caching, served through a modular Node.js/PostgreSQL REST backend with independently deployable Docker services

EDUCATION

University of Texas at Austin | M.S. in Computer Science

Aug 2026 - May 2028 (Expected)

Relevant Coursework: Algorithms, Machine Learning, Advanced OS, Distributed Computing, Deep Learning, Parallel Systems, NLP

Texas A&M University | B.S. in Applied Mathematics and Computer Science | Minor: Statistics | GPA: 3.8

December 2025

Relevant Coursework: DSA, Discrete Structures, Machine Learning, Algorithm Analysis, Computer Architecture

Publications: Yarram et al., AERA 2026, Los Angeles, CA — computational modeling and STEM+C mentorship research [Poster session]

SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, MySQL, Bash

Frameworks/Libraries: ReactJS, Spring Boot, Pandas, GraphQL, TensorFlow, scikit-learn, PyTorch, LangChain, MCP, RAG pipelines

Technologies: Docker, Kubernetes, NGINX, Kafka, AWS SageMaker, Azure, Elasticsearch, FastAPI, Snowflake, Redis, Grafana, Splunk

Methodologies: REST APIs, API gateway design, mTLS/PKI, application security, Agile/Scrum, OOP/OOD, CI/CD, distributed systems, Git

Certifications: Oracle Certified Professional: Java SE 11 Developer; Supervised ML: Regression and Classification